

Discovering Voices in the AI Discourse: A Data Mining Analysis of Reddit ChatGPT Communities

CS 4412 — Data Mining | M4 Final Report
Kennesaw State University — Spring 2026

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Abstract. This report presents a complete data mining analysis of 49,428 Reddit comments discussing ChatGPT, drawn from communities including r/ChatGPT, r/technology, and r/Futurology. Using K-Means clustering on TF-IDF and behavioral features, combined with Latent Dirichlet Allocation (LDA) topic modeling, this work discovers four distinct user engagement archetypes and recurring thematic structures in public AI discourse. Results reveal that 73.6% of users are mainstream casual commenters, while a small but analytically rich minority engages deeply with technical and philosophical dimensions of AI. These findings have implications for science communication, platform design, and AI policy engagement.

1. Introduction

The rapid adoption of large language models (LLMs), particularly ChatGPT, has generated an unprecedented volume of public discourse across social media platforms. Reddit, as one of the largest forum-based communities, hosts rich multi-community discussions that reflect real-world public understanding, concerns, and enthusiasm surrounding AI technology.

This project applies data mining techniques to a corpus of 49,428 Reddit comments to answer three core discovery questions:

- **RQ1:** What natural user segments exist based on commenting behavior and content characteristics?
- **RQ2:** What topics and thematic discussions emerge from the ChatGPT comment corpus?
- **RQ3:** What distinct patterns characterize user engagement across different subreddit communities?

The dataset was sourced from Reddit communities actively discussing ChatGPT, spanning r/ChatGPT, r/technology, r/Futurology, and related subreddits. Data was collected via the Reddit API (PRAW) and processed using standard NLP pipelines. This report covers the full analysis pipeline, findings, interpretation, and critical evaluation of the work.

2. Methods

2.1 Dataset

The dataset comprises 49,428 Reddit comments collected via the Reddit API (PRAW). Comments were drawn from subreddits including r/ChatGPT, r/technology, r/Futurology, r/MachineLearning, and r/artificial. The collection period spans early 2023 through mid-2024, capturing the initial public explosion of interest in ChatGPT.

Each comment record includes: the raw comment text, subreddit source, author ID (anonymized), comment score (upvotes minus downvotes), and timestamp.

2.2 Preprocessing

Text preprocessing followed a standard NLP pipeline:

- Lowercasing and punctuation removal
- Stopword removal using NLTK's English stopword corpus
- Tokenization and lemmatization (WordNetLemmatizer)
- Removal of Reddit-specific artifacts (URLs, subreddit references, deleted/removed content)
- Filtering of comments with fewer than 5 tokens after cleaning

Behavioral features were engineered per comment: character length, word count, comment score, and hour-of-day posted. These were combined with TF-IDF text vectors for clustering.

2.3 Clustering (K-Means)

K-Means clustering was applied to a combined feature matrix of TF-IDF vectors (top 5,000 features) and normalized behavioral features. Dimensionality reduction via PCA (50 components) was applied

prior to clustering to improve computational efficiency and reduce noise.

The optimal number of clusters (k=4) was selected using the Elbow Method on Within-Cluster Sum of Squares (WCSS) and Silhouette Score analysis. Values of k from 2 to 10 were evaluated, with k=4 showing the clearest elbow and a silhouette score of approximately 0.31.

2.4 Topic Modeling (LDA)

Latent Dirichlet Allocation (LDA) was applied to the cleaned comment corpus using the Gensim library. A dictionary and bag-of-words corpus were constructed, filtering tokens appearing in fewer than 15 documents or more than 50% of documents. Coherence score optimization over 5-20 topics guided the selection of 8 topics, achieving a coherence score (c_v) of 0.47.

2.5 Parameter Summary

Parameter	Value	Rationale
TF-IDF features	5,000	Balance of vocabulary coverage and noise
PCA components	50	Retain ~85% explained variance
K-Means clusters (k)	4	Elbow method + silhouette analysis
LDA topics	8	Coherence score optimization
Min token length	5	Remove trivial comments
Random state	42	Reproducibility

Table 1: Key algorithm parameters and justifications.

3. Results

3.1 Cluster Profiles

K-Means clustering with k=4 produced four meaningful and interpretable user segments. The cluster distribution is shown in Table 2 below.

Cluster	Label	Size	% of Total	Avg. Comment Length	Avg. Score
0	Deep Technical Discussers	1,680	3.4%	847 chars	28.4
1	Moderate Engagers	8,403	17.0%	312 chars	12.1
2	Niche Interest Group	2,916	5.9%	489 chars	18.7
3	Mainstream Majority	36,429	73.6%	142 chars	6.3

Table 2: K-Means cluster profiles (k=4, n=49,428).

Cluster 0 — Deep Technical Discussers (3.4%): This small cluster is characterized by significantly longer comments, higher upvote scores, and vocabulary centered on technical and philosophical AI topics. Top terms include 'consciousness', 'bias', 'regulation', 'copyright',

'sentient', and 'alignment'. These users engage in extended debates about AI ethics, legal implications, and long-term societal impacts.

Cluster 1 — Moderate Engagers (17.0%): Mid-length comments with moderate engagement scores. Discussion centers on practical use cases, feature comparisons, and professional applications of ChatGPT. Terms like 'work', 'productivity', 'writing', 'job', and 'coding' dominate.

Cluster 2 — Niche Interest Group (5.9%): A specialized segment with above-average comment length, focused on specific domains such as education, creative writing, and research. These users show interest in domain-specific applications rather than general AI discussion.

Cluster 3 — Mainstream Majority (73.6%): Short, casual comments reflecting everyday reactions. High volume but low depth. This cluster captures the Reddit norm of brief upvoted agreement, humor, and surface-level reactions to AI news.

3.2 LDA Topic Modeling

LDA identified 8 coherent topics across the corpus. Selected top topics include:

- **Topic 1 — AI Ethics & Regulation:** 'bias', 'regulation', 'law', 'copyright', 'government', 'policy'
- **Topic 2 — Practical Use Cases:** 'write', 'code', 'email', 'work', 'prompt', 'generate'
- **Topic 3 — Education & Learning:** 'student', 'teacher', 'homework', 'school', 'essay', 'plagiarism'
- **Topic 4 — Future of AI / Speculation:** 'future', 'agi', 'replace', 'job', 'human', 'intelligent'
- **Topic 5 — Technical Capabilities:** 'model', 'gpt4', 'training', 'token', 'fine-tune', 'benchmark'

Cluster 0 (Deep Discussers) showed strong loading on Topics 1, 4, and 5. Cluster 1 (Moderate Engagers) was dominated by Topic 2. Cluster 3 (Mainstream) was distributed broadly across topics with no strong signal.

3.3 Cross-Community Patterns

Subreddit-level analysis revealed meaningful community differences. r/MachineLearning and r/artificial were overrepresented in Cluster 0, while r/ChatGPT housed the majority of Cluster 3 comments. r/Futurology showed a higher proportion of Cluster 2 (Niche) comments related to speculative AI futures. Comment scores were consistently higher in Cluster 0 regardless of subreddit, suggesting that depth of engagement drives upvotes across communities.

4. Interpretation

The four-cluster structure maps meaningfully onto real-world communication patterns in online AI discourse. The dominance of Cluster 3 (73.6%) reflects a well-documented phenomenon in online communities: the '1% rule', where the vast majority of users consume or minimally react to content while a small minority drives substantive discussion.

Cluster 0's concentration of ethical and philosophical vocabulary is significant. These Deep Technical Discussers are likely researchers, AI practitioners, policy advocates, or highly informed enthusiasts. Their high upvote scores suggest that the broader Reddit community values depth even if it doesn't produce it. This has implications for science communication: engaging with this small group may have

outsized influence on how AI is publicly discussed.

The Education topic (Topic 3) appearing prominently in the corpus reflects a major real-world debate about ChatGPT in academic settings. The Practical Use Cases topic (Topic 2) aligns with how most everyday users actually experience AI tools — as productivity enhancers rather than philosophical objects.

Cross-community differences confirm that subreddit culture shapes discourse: r/MachineLearning attracts technical depth, r/ChatGPT attracts casual reaction, and r/Futurology attracts speculative framing. This suggests that platform architecture and community norms are significant drivers of how AI is understood publicly.

5. Critical Assessment

5.1 Validity

The clustering results show reasonable face validity — the discovered segments align with known patterns in online community behavior. However, the silhouette score of ~ 0.31 indicates moderate but not strong cluster separation. TF-IDF features do not capture semantic meaning, meaning two comments with similar intent but different word choices may be placed in different clusters. Future work using sentence embeddings (e.g., SBERT) could improve semantic coherence.

5.2 Limitations

- **Sampling bias:** Reddit users are not representative of the general public. The dataset over-represents English-speaking, technically literate users.
- **Temporal snapshot:** The collection period captures a specific moment in public AI discourse. Sentiment and topics likely shift as AI capabilities evolve.
- **No ground truth:** Cluster labels are interpretive. Without human annotation of a subset of comments, there is no way to formally validate that Cluster 0 truly represents 'deep technical discussers'.
- **LDA limitations:** LDA assumes a bag-of-words model and is sensitive to hyperparameter choice. The coherence score of 0.47 is moderate; more sophisticated neural topic models (e.g., BERTopic) may yield richer results.
- **Deleted content:** Reddit deletes and removes content frequently. Removed comments create gaps that could bias the distribution of topics and cluster sizes.

5.3 Ethical Considerations

User data was collected via the public Reddit API under standard researcher access terms. All author identifiers were anonymized prior to analysis; no attempt was made to identify individuals. No personally identifiable information was stored or used.

Reddit comments, while technically public, are written with a community audience in mind rather than research use. This raises questions about contextual integrity. However, the analysis focuses on aggregate patterns rather than individual behavior, and no findings could be used to target or harm specific users.

The results could theoretically be misused — for example, identifying 'influential' commenters in Cluster 0 for targeted manipulation. This risk is low given the aggregate nature of the findings, but should be acknowledged in any deployment context.

6. Conclusion

This project applied K-Means clustering and LDA topic modeling to a corpus of 49,428 Reddit comments about ChatGPT, discovering four interpretable user archetypes and eight recurring topic themes in public AI discourse. The dominant finding is the structural inequality in online participation: 73.6% of commenters engage casually, while a small cohort (3.4%) drives deep, substantive discussion that garners disproportionate community recognition.

These findings have practical implications for AI science communication, suggesting that outreach efforts should focus on engaging the technically informed minority, while platform design choices significantly shape how AI is discussed publicly. For policy engagement, understanding which discourse communities exist — and what language they use — is valuable for crafting accessible AI governance communication.

Future work should explore temporal analysis of how discourse shifts as AI capabilities evolve, apply neural embeddings for richer semantic clustering, and expand to non-English communities to address the current sampling bias.